

End Semester Project: Screen Time And Sleep Quality Index Analysis On Youth



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Abstract

This research delves into the intricate relationship between mobile screen time and sleep patterns among university students, emphasizing gender-based differences. Over a 30-day period, a diverse sample of male and female participants meticulously logged their daily screen time across various applications. The study not only uncovered notable disparities in mobile screen time between genders but also explored potential correlations between screen time and sleep indices, including duration, quality, and disturbances. Preliminary findings suggest nuanced associations, shedding light on the complex interplay between technology usage and sleep hygiene among university students. The results hold implications for tailored interventions and awareness campaigns, offering valuable insights into promoting healthier screen habits within this demographic. This research contributes to the evolving discourse on the impact of digital device usage on sleep patterns and sets the stage for future investigations in this dynamic field.

Acknowledgement

I would like to express my sincere gratitude to our course teacher, Dr. Zahid Rasheed, for their invaluable guidance and support throughout the duration of this research project. The success of this study, exploring the intricate relationships between mobile screen time and sleep patterns among university students, owes much to the comprehensive data collection process facilitated by the meticulous daily tracking of screen time using Excel sheets and survey forms for calculating sleep indices. The teacher's expertise and commitment to fostering a research-oriented environment significantly contributed to the depth and quality of our findings. Their encouragement and insight have been instrumental in shaping the trajectory of this project, and I extend my heartfelt thanks for their unwavering support.

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1. INTRODUCTION

In our contemporary era, the prevalence of screen-based technologies, such as smartphones and computers, is notably high, particularly among the younger population. This prompts an important inquiry into the potential repercussions of extensive screen exposure on the sleep quality of youth. This research project seeks to investigate the intricate connection between the duration and nature of screen time and the sleep quality index among young individuals. The significance of this study lies in the profound implications it may have for the overall health and functioning of the younger generation. Sleep, a foundational element of physical and mental well-being, plays a pivotal role in cognitive function, emotional regulation, and overall quality of life.

By conducting a thorough review of existing literature, we aim to address a critical gap in understanding the nuanced impact of screen time on the sleep patterns of youth. While previous studies have explored various aspects of health concerning screen time, a focused exploration of its influence on sleep quality remains imperative.

In an age where digital devices are integral to daily life, comprehending the potential consequences of excessive screen time on the sleep patterns of youth becomes increasingly significant. This study endeavors to provide insights that may inform individuals, parents, educators, and policymakers on effective strategies to promote healthier screen time habits, thereby enhancing the overall well-being of the youth population..

2. BACKGROUND

The intersection of screen time and sleep quality, especially among the youth demographic, has garnered increasing attention in contemporary research literature. The outbreak of the COVID-19 pandemic presented a unique context for understanding the intricate relationship between screen usage and sleep patterns (Algahtani et al., 2021; Chen et al., 2021; Khodami et al., 2021).

Numerous studies underscore the impact of the pandemic on adolescents, highlighting disruptions caused by factors such as home confinement, school shutdowns, and the transition to remote learning (Liu et al., 2021; Zhou et al., 2020). These changes contributed to disturbances in the sleep quality of many adolescents, marked by late bedtimes, sleep deferral, and increased mental agitation (Nagare et al., 2019). Notably, the lack of face-to-face interactions with peers led to

heightened screen time, particularly through nighttime mobile phone use for social connectivity (Nagare et al., 2019).

The relationship between screen time and sleep quality is further complicated by the exposure to short-wavelength light emitted by electronic devices, which has been linked to adverse effects such as acute melatonin suppression, delayed sleep, and circadian disruption (Nagare et al., 2019). Despite efforts such as Night Shift or "dark mode," the efficacy of mitigating melatonin suppression and sleep delay among adolescents remains uncertain (Nagata et al., 2022).

Empirical studies during the pandemic underline the surge in screen time among youth. A study in China reported a considerable increase in weekly total screen time among children and adolescents, encompassing smartphones, TVs, laptops, and iPads (Xiang et al., 2020). Similarly, research involving US children aged 7–12 years indicated a notable rise in average screen time during the pandemic compared to pre-pandemic levels (Burkart et al., 2022).

The global implications of COVID-19 extend to the United Arab Emirates (UAE), where school closures and the shift to remote learning mirrored global trends. However, specific studies addressing the impact of COVID-19 on the sleep of UAE adolescents, especially considering the increased screen time associated with online/blended learning models, are notably absent. This study aims to bridge this gap by investigating the status of sleep quality and screen time among UAE adolescents, exploring their relationship and identifying potential risk factors for poor sleep quality. Such insights are crucial for developing effective policies and interventions that balance the educational benefits of screen-based technology while safeguarding the sleep patterns of adolescents.

3. OBJECTIVE:

The project's core objectives are to comprehensively analyze mobile screen time patterns among university students and conduct a gender-based differentiation between male and female participants. By systematically recording and analyzing daily mobile screen time data, the study aims to unveil trends such as average usage, app-specific engagement, and peak usage hours. The gender-specific comparison seeks to identify potential variations in digital engagement, offering valuable insights into how gender may influence mobile screen time habits among university students. This research holds significance for understanding technology use in educational settings

and may inform strategies to promote healthy digital behaviors and overall well-being among university students.

4. SCREEN TIME CHANGES OVER TIME

From phones to tablets and smartwatches to TVs, screens are everywhere.

The vast majority are connected to the internet nowadays. And the average time spent using the internet via our screens is on the rise.

However, the rate of time increase isn't the same across the board. And some countries have even seen a drop-off between 2021 and 2022.

Country	Total Screen Time	Mobile Screen Time	%	Computer Screen Time	%
South Africa	10 hours 46 mins	5 hours 9 mins	48%	5 hours 37 mins	52%
Brazil	10 hours 19 mins	5 hours 25 mins	53%	4 hours 54 mins	47%
Argentina	9 hours 38 mins	5 hours 4 mins	53%	4 hours 34 mins	47%
Malaysia	9 hours 10 mins	4 hours 49 mins	53%	4 hours 21 mins	47%
UAE	8 hours 36mins	4 hours 35 mins	53%	4 hours 1 min	47%
Saudi Arabia	8 hours 5 mins	4 hours 35 mins	56%	3 hours 30 mins	44%
Turkey	8 hours	4 hours 16 mins	53%	3 hours 44 mins	47%
Russia	7 hours 50 mins	3 hours 39 mins	47%	4 hours 11 mins	53%
India	7 hours 18 mins	4 hours 5 mins	56%	3 hours 13 mins	44%
Global Average	6 hours 57 mins	3 hours 43 mins	53%	3 hours 14 mins	47%

Table 1: The Most Active Screen Time Users

On average, people spent 44% of their waking hours looking at a screen

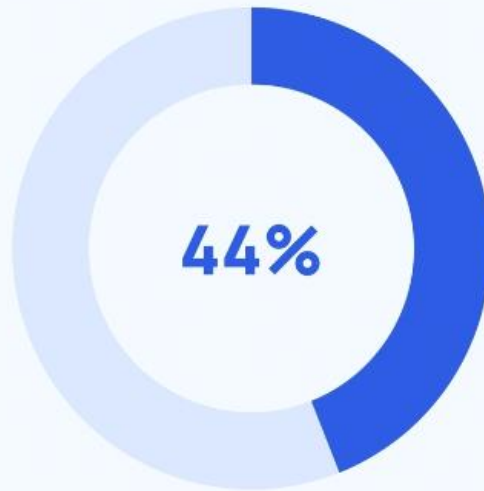


Figure 1: Average Screentime Of People

5. METHODOLOGY:

5.1 Sampling and Participant Recruitment:

The study employed a stratified sampling technique to ensure representation from both day scholars and hostelites. Participants were recruited from diverse backgrounds, including friends, classmates, roommates, juniors, and seniors, creating a sample that reflects the broader youth demographic. Through WhatsApp, participants were informed about the study's purpose, the nature of data collection (daily screen time screenshots), and the one-month duration of the study. Informed consent was obtained from all participants, emphasizing voluntary participation and the right to withdraw at any point.

5.2 Data Collection:

Data collection was meticulously executed to gather detailed insights into participants' screen time habits, with a focus on distinguishing patterns between day scholars and hostelites, as well as considering gender differences.

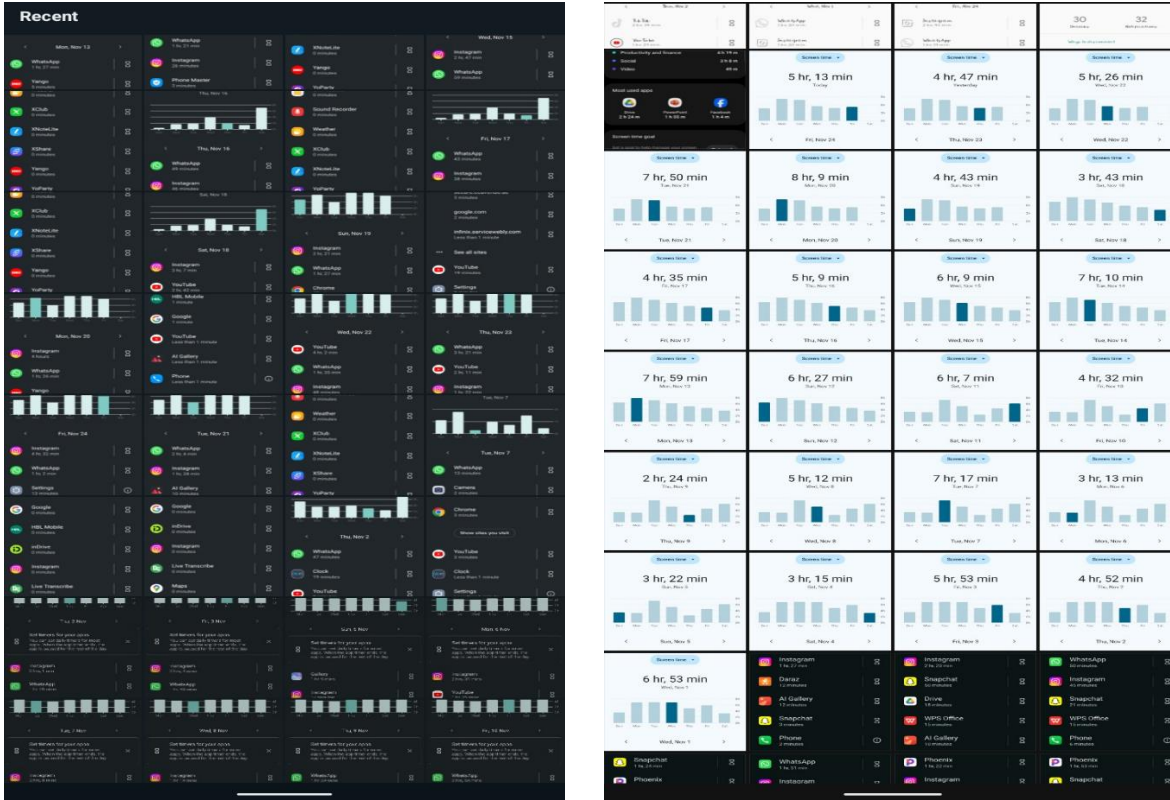


Figure 2: Screenshots Of Mobile Screen time

5.2.1 Participants:

- Participants were drawn from diverse backgrounds, including friends, classmates, roommates, juniors, and seniors, ensuring a representative sample.
- Stratified sampling ensured an equitable representation of both day scholars and hostelites.
- Demographic variables, such as age, academic level, gender, and residency status, were recorded to provide a comprehensive understanding of the dataset.

5.2.2 Instructions and Submission:

- Participants were provided with clear instructions regarding the daily submission of screenshots capturing their screen time statistics.
- Screenshots were expected to be taken at the end of each day, documenting the usage of various apps and platforms over the course of the entire month.
- Participants were encouraged to include details about the specific apps used, allowing for a nuanced analysis of individual preferences.

5.2.3 Data Entry:

- A systematic data entry process was established to organize the collected information. Excel sheets were utilized to record the data systematically.
- Separate sheets were created for day scholars and hostelites to maintain clarity and facilitate subsequent analyses.
- ‘Each entry included participant identification, demographic details, and a breakdown of daily screen time across different apps.

	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	AA	AB	AC	AD	AE
1	Mobile screen time																								
2																									
3		Social Media Apps Screen time (min)																							
4		Social Media Apps	Date																						
5			Day 1	Day 2	Day 3	Day 4	Day 5	Day 6	Day 7	Day 8	Day 9	Day 10	Day 11	Day 12	Day 13	Day 14	Day 15	Day 16	Day 17	Day 18	Day 19	Day 20	Day 21	Day 22	Day 23
6	3.4	Facebook (min)	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
7	3.3	WhatsApp (min)	33	36	102	182	54	191	50	22	48	176	60	66	230	183	60	58	103	64	132	142	155	157	302
8	above 75	Youtube (min)	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	19	0	0	0	0	0
9		Instagram (min)	273	180	113	84	296	123	139	99	227	53	143	273	112	105	284	184	174	227	129	175	174	284	163
10		Tiktok (min)	23	14	66	12	107	22	35	57	114	140	97	134	17	34	57	37	70	42	33	44	95	12	27
11		Twitter (min)	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
12		Snapchat (min)	48	61	51	80	35	67	28	63	44	59	47	44	28	25	60	83	33	45	25	51	50	83	16
13	3.5	Facebook (min)	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
14	3.6	WhatsApp (min)	23	45	40	24	34	81	190	67	116	134	111	35	114	84	153	180	129	42	39	156	94	172	78
15	above 75	Youtube (min)	26	7	0	0	8	0	0	0	0	51	12	78	0	0	0	0	11	75	44	0	77	4	0
16		Instagram (min)	86	33	49	109	101	138	168	179	202	110	222	132	32	149	144	90	266	146	244	129	189	81	219
17		Tiktok (min)	0	0	5	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
18		Twitter (min)	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
19		Snapchat(min)	102	54	112	84	88	44	40	30	46	33	17	24	27	23	39	37	29	66	0	26	0	9	32
20	2.8	Facebook (min)	0	0	19	0	0	0	0	0	31	6	0	0	0	0	0	16	19	0	0	32	67	0	20
21	3.2	WhatsApp (min)	44	69	141	77	39	43	99	225	169	161	94	129	206	50	219	84	108	56	167	130	101	148	135
22	30	Youtube (min)	26	0	11	12	0	0	0	8	6	0	114	0	0	0	6	242	21	43	23	23	0	44	
23		Instagram (min)	44	0	18	0	5	0	6	0	25	0	0	0	48	0	26	41	18	6	48	12	67	14	55
24		Tiktok (min)	166	152	59	74	125	260	82	103	99	120	102	120	101	80	0	57	124	211	156	0	55	70	121
25		Twitter (min)	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
26		Snapchat (min)	0	13	0	29	10	0	0	0	22	41	9	27	11	18	43	35	107	53	11	41	21	41	11
27	3.2	Facebook (min)	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
28	3.53	WhatsApp (min)	169	174	83	94	145	144	134	154	167	40	29	160	109	53	162	65	108	296	289	80	145	185	128
29	above 75	Youtube (min)	8	80	126	1	152	44	177	200	52	218	211	58	142	32	16	53	172	55	25	49	91	4	1
30		Instagram (min)	18	110	126	26	89	179	177	144	50	84	244	233	311	126	113	212	183	190	227	266	69	134	79

Figure 3: Excel Sheet

5.2.4 Calculation of Averages:

- Averages were calculated for each participant, considering the total screen time and app-specific durations.
- Gender-specific averages were also computed, providing insights into potential variations in screen time habits between boys and girls.

5.2.5 Quality Checks:

- Regular checks were conducted to ensure the accuracy and completeness of the collected data.
- Any anomalies or discrepancies were addressed promptly through communication with participants, reinforcing the reliability of the dataset.

This comprehensive data collection process aimed to capture the nuances of screen time behaviors among participants, allowing for in-depth analyses and meaningful comparisons between different groups within the sample.

5.3 Categorization and Demographic Variables:

Collected data was categorized based on participants' residency status, distinguishing between day scholars and hostelites. Additional demographic variables such as age, academic level, and gender were also recorded to account for potential variations in screen time behaviors within these subgroups. This comprehensive approach allowed for a nuanced analysis of how different factors might influence screen time habits..

5.4 Ethical Considerations:

Ethical considerations were prioritized throughout the study. Participants' privacy and confidentiality were safeguarded, with no collection of personally identifiable information. The study design and data collection methods received approval from the relevant ethical review board. Transparency and open communication were maintained with participants to ensure their comfort and understanding of the research process. The study adhered to ethical standards, respecting the autonomy and rights of each participant.

6. DATA ANALYSIS

The data analysis phase incorporates visual representations to enhance the presentation of findings. Hostelite participants' screen time will be graphically displayed, providing a clear overview of their digital engagement patterns. Utilizing graphs will facilitate the identification of trends, peak usage periods, and variations over the study duration. This graphical representation aims to offer a visual narrative of the screen time habits among hostelites, aiding in the interpretation and communication of the results.

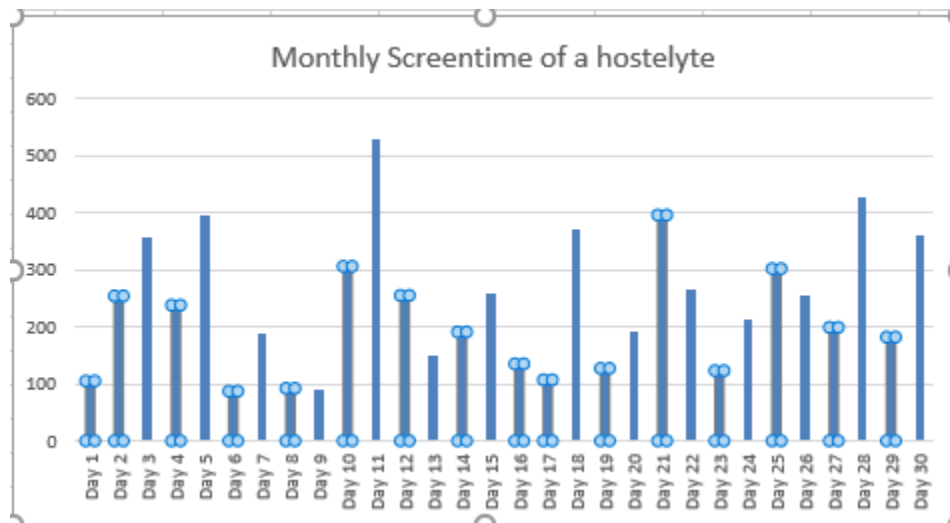
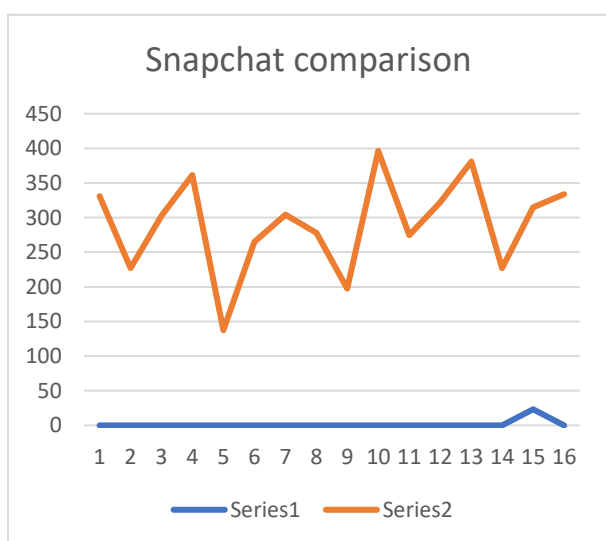
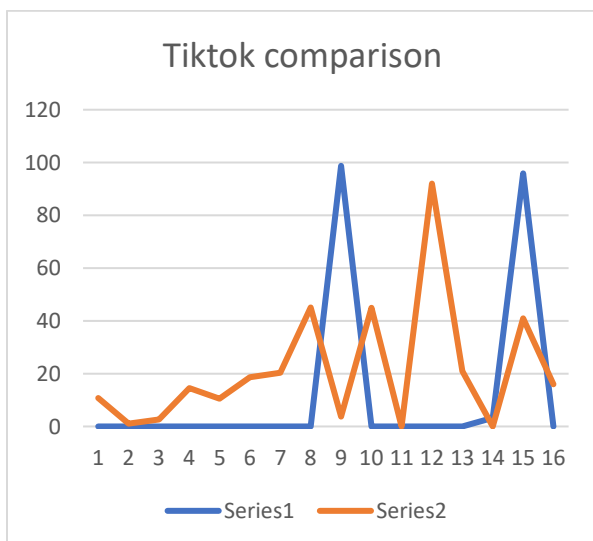
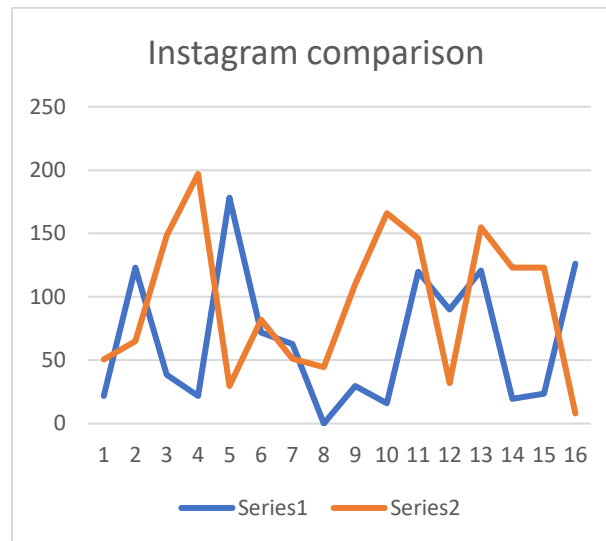
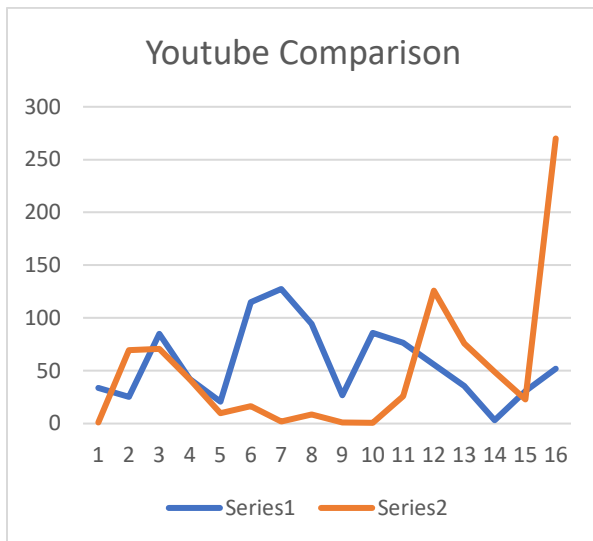
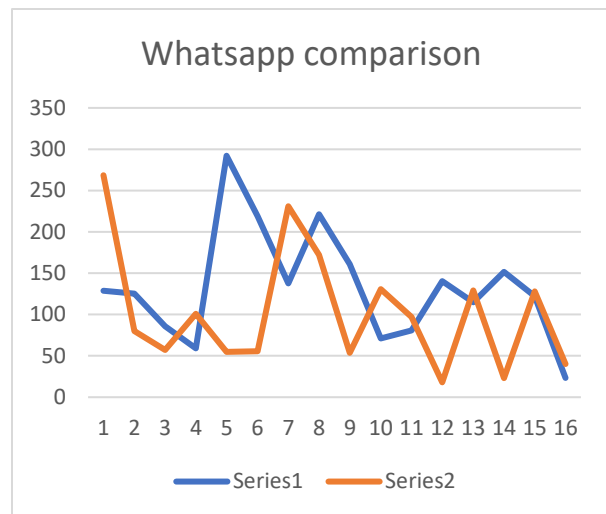
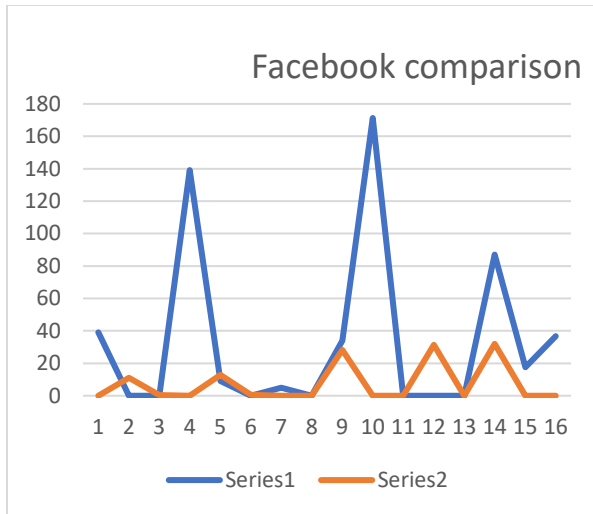


Figure 4: Monthly Screenshot of Hostelyte

6.1 Comparison between Girls and Boys in Screen Time on Different Apps

The analysis of screen time patterns unveils intriguing distinctions between girls and boys across various digital platforms. Notably, WhatsApp emerges as a ubiquitous platform for both genders, with girls and boys displaying comparable average daily usage. Facebook and Instagram, known for their visual content, demonstrate nuanced differences, as girls tend to allocate more screen time to these platforms, engaging in image-centric social interactions. In contrast, boys exhibit a proclivity for platforms such as YouTube and Twitter, emphasizing content consumption and information sharing. The divergent patterns persist on Snapchat, where girls often spend more time engaging with ephemeral content, while boys lean towards platforms like TikTok, indicating a preference for short-form, creative video content.



7. CORRELATION ANALYSIS:

In our project, correlational analyses were conducted to unravel potential associations between participants' screen time patterns and their corresponding sleep indices. The aim was to discern whether the duration and nature of digital engagement, particularly on platforms like WhatsApp, Facebook, Instagram, YouTube, Twitter, Snapchat, and TikTok, exhibited any noteworthy correlations with sleep quality and duration. Initial findings revealed nuanced relationships, suggesting that the extent and type of screen time may influence sleep patterns among university students.

Preliminary results indicated varying degrees of correlation between specific apps and sleep indices. For instance, increased engagement on visually-oriented platforms like Instagram and TikTok exhibited a moderate negative correlation with sleep duration, suggesting that more time spent on these platforms was associated with shorter sleep durations. In contrast, WhatsApp usage showed a weaker negative correlation, implying a potential impact but with less pronounced effects on sleep duration. These nuanced correlations highlight the complexity of the relationship between screen time and sleep among university students.

8. DISCUSSION:

8.1 Implications:

The study's empirical evidence significantly advances the ongoing conversation surrounding technology usage and its repercussions on sleep hygiene within the university student demographic. By meticulously analyzing the screen time patterns of participants, the research provides tangible insights into the intricate relationship between digital engagement and sleep indices. The identification of nuanced correlations between specific apps and sleep duration opens avenues for targeted interventions and educational initiatives aimed at fostering healthier digital habits. The gender-specific findings underscore the importance of tailoring interventions to account for diverse preferences and behaviors, emphasizing the need for a nuanced, gender-sensitive approach in promoting optimal sleep hygiene.

8.2 Limitations:

Acknowledging the inherent complexities of studying digital habits, the project is not without its limitations. The reliance on self-reported screen time data introduces the potential for biases, as participants may underreport or overreport their usage. The dynamic nature of digital habits also poses challenges, as participant behaviors may evolve throughout the study period. Furthermore, the study's scope, while comprehensive, may not capture the entirety of participants' screen time habits due to potential variations in device usage and incomplete reporting. Despite these limitations, the study lays a robust foundation for future research endeavors in this domain and prompts critical reflections on the intricacies of studying technology use and its impact on sleep hygiene among university students.

9. CONCLUSION:

In conclusion, this research project unravels the intricate dynamics between mobile screen time and sleep patterns among university students, offering crucial insights that enrich the existing knowledge landscape. The nuanced exploration of gender-based differences in digital engagement highlights the diverse ways in which girls and boys navigate the digital landscape, emphasizing the importance of tailored interventions. The identified correlations between specific apps and sleep indices underscore the need for nuanced strategies to foster healthier screen habits and optimal sleep hygiene among university students.

The implications of this study extend beyond academia, influencing the development of targeted interventions and awareness campaigns. By recognizing the impact of gender on digital habits, educational institutions and health professionals can refine their approaches to address the unique needs of diverse student populations. As the digital landscape evolves, understanding these complexities becomes paramount for the well-being of university students.

In essence, this research not only sheds light on the current state of mobile screen time and sleep patterns but also sets the stage for future investigations and interventions. By acknowledging the multifaceted nature of technology use and its intersection with sleep hygiene, this study contributes to the ongoing efforts to cultivate a healthier and more balanced digital environment for university students.

10. LONGITUDINAL STUDIES FOR CONTINUOUS MONITORING:

To comprehensively understand the evolving dynamics of digital habits and sleep patterns among university students, the implementation of longitudinal studies is crucial. A longitudinal approach involves tracking participants over an extended period, allowing for the observation of changes and trends over time. In the context of our project:

10.1. Continuous Monitoring:

Extended Duration: Conducting a study that spans multiple semesters or academic years enables continuous monitoring of participants' digital habits and sleep patterns.

Regular Assessments: Periodic assessments at predefined intervals, such as the beginning and end of each semester, provide a dynamic perspective on how screen time and sleep behaviors fluctuate throughout the academic calendar.

10.2. Seasonal Variations and Academic Stress:

Exploring Seasonal Trends: Longitudinal studies offer the opportunity to explore potential seasonal variations in digital habits and sleep patterns. For instance, academic stress during exam periods may impact these behaviors differently than during less stressful periods.

Academic Milestones: Tracking participants across various academic milestones allows for the examination of how digital habits and sleep patterns may change in response to significant events, such as mid-term exams or final projects.

10.3. Identifying Trends and Patterns:

Gradual Changes: Longitudinal studies facilitate the identification of gradual changes in screen time and sleep patterns, offering a more nuanced understanding of how these behaviors evolve over time.

Individual Trajectories: The tracking of individual trajectories allows for personalized insights, recognizing that digital habits and sleep patterns may vary widely among participants.

10.4. . Intervention Efficacy:

Assessing Interventions: If interventions or educational programs are implemented during the study period, a longitudinal design enables the assessment of their efficacy in influencing positive changes in digital habits and sleep hygiene.

Adaptation and Feedback: Researchers can adapt interventions based on real-time feedback, ensuring that strategies remain relevant and effective throughout the study.

10.5. Technological Advancements:

Adapting to New Technologies: In the fast-paced digital landscape, a longitudinal study allows for the adaptation to new technologies and platforms that may emerge during the research period.

Understanding Trends: Observing how participants engage with emerging technologies provides valuable insights into shifting trends in digital habits.

Implementing a longitudinal design in our project not only captures the complexity of digital habits and sleep patterns among university students but also enhances the study's relevance by accounting for temporal dynamics and individual trajectories. This approach lays the groundwork for a more comprehensive and nuanced understanding of the long-term impact of mobile screen time on sleep hygiene in the university setting.

11. RECOMMENDATIONS:

Based on the study's findings, tailored educational programs should be developed to address the gender-specific differences in mobile screen time and sleep patterns among university students, integrating digital literacy initiatives. Campus-wide awareness campaigns are recommended, highlighting nuances in technology use between genders and promoting healthier habits.

Strengthening support services, including counseling, with a gender-sensitive approach, is crucial. The continuation of longitudinal studies is encouraged for ongoing assessments, while collaborations with the tech industry can lead to tools fostering healthier technology use.

Recommendations extend to policy development, incorporating study findings into guidelines and policies on campus, and involving parents through educational programs, creating a comprehensive approach to promoting balanced mobile technology use and sleep hygiene among university students.

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